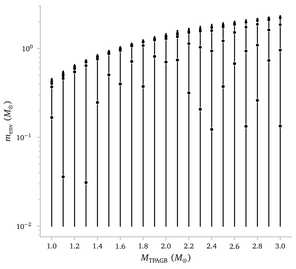
Koen Essers • Master Internship • November 4, 2026

Notes Week 35

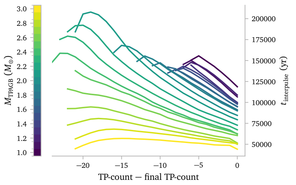
a. Different Masses

I want to see if there is some predictable structure in the plot below:



The dots represent the mass of the TPAGB star at the start of a certain thermal pulse. The exact thermal pulse numbers are irrelevant here. It is only important to look at the lowest two dot. These dot represent the mass of the envelope during the *last* two thermal pulse. Clearly, if we interact during the last *interpulse* phase, RLOF will occur when the star has an envelope mass between these two values.

The figure above shows that for the models that interact during the interpulse, the envelope mass is approximately 0.5 $M_\odot$. This figure shows that this is totally in agreement with the values that we expect are possible. The envelope mass of the 1.4 $M_\odot$ star during the last thermal pulse is not very low compared to the other stars.



During the last thermal pulses, there is quite a strong correlation between the mass and the interpulse duration. However (unfortunately), during the earlier thermal pulses, this relation is a bit less clear.

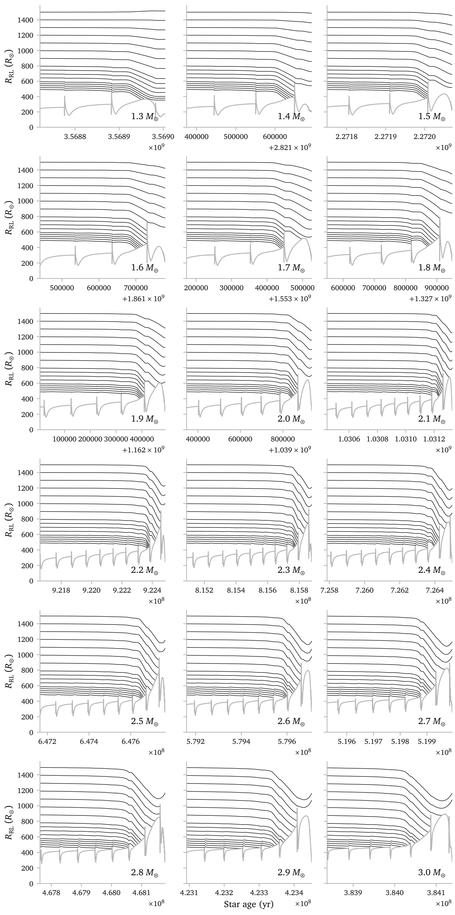
Unfortunately, the changing interpulse duration *already* makes it difficult to predict the mass of the envelope during the thermal pulses. This is even without taking into account that the wind mass loss is also not constant throughout the evolution.

I am interested to see how drastically the envelope mass changes between masses. The first figure makes it seem like it is very chaotic. I want to simulate some more single stars between some masses to see if the chaotic nature is really part of the TP-AGB evolution, *or*, if it is just an effect of the rough sampling or 0.1 $M_\odot$ I am using.

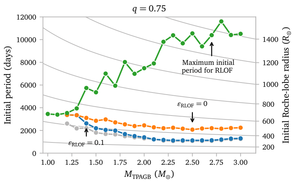
I choose to simulate the masses 1.9 to 2.1 with a delta of 0.01 $M_\odot$. In the first figure, we see that these masses have roughly the same envelope mass during the final thermal pulse. However, they all have a different amount of thermal pulses, so the evolution *must* be quite different.

b. Effects of Different Masses

To start, I generate a new grid with a dense mass and initial Roche lobe grid. Below, I show the predicted simple binary tracks.

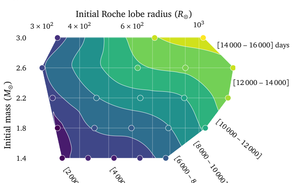


I also do some basic binary searches to find the approximate initial Roche lobe radii / periods for models that experience RLOF and could produce a barium star.



Clearly, the minimum initial period does not change much with mass, while the maximum initial period does. This makes sense, as the final radii of heavier TPAGB stars is *much* larger.

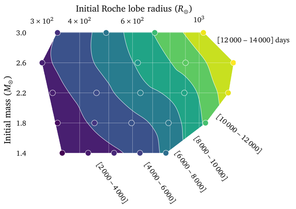
Below, I show some preliminary results based on the coarse grid of different masses that I simulated.



This plot shows the final period of a grid of models with $q = 0.85$, $\epsilon = 0.1$ and $f_{beta} = 0.1$.

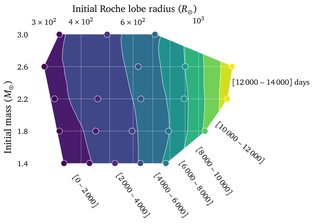
Unfortunately, the resolution of the models in the M_{TPAGB}-R_{RL} plane is not great. I color the plane using cubic interpolation between the datapoints. The contours are definitely not precise, and not too much importance should be attributed to the contours.

These models have the *lowest* f_β values. This means these models have the *largest* periods.



This plot shows the final period of a grid of models with $q = 0.85$, $\epsilon = 0.1$ and $f_{beta} = 0.2$.

As expected, the higher value of $f_\beta = 0.2$ produces lower final periods. However, there is also a slightly different effect that can be seen when comparing this plot with the one above: The mass becomes less important, while the initial Roche lobe radius stays relevant.

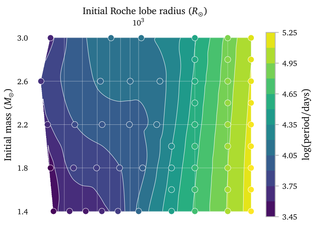


This plot shows the final period of a grid of models with $q = 0.85$, $\varepsilon = 0.1$ and $f_{\mathrm{beta}} = 0.3$.

The same trends are visible for the models with $f_\varepsilon = 0.3$. Now, the mass *really* seems like it is not relevant anymore.

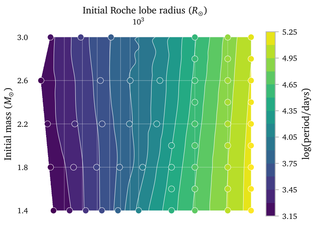
Obviously, things would change if we would take into account the models that do not undergo RLOF.

Let me show how these wind only models would complete the plane on the right.



This plot shows a much broader plane of the final period of a grid of models with $q = 0.85$, $\varepsilon = 0.1$ and $f_{\mathrm{beta}} = 0.1$.

Clearly the models on the right follow quite a different trend then the RLOF models. For the wind models, we see that the final period follows the same trend as the initial period, where the longest periods correspond to the *largest* initial Roche lobe radius and the *least heavy* donor stars.



This plot shows a much broader plane of the final period of a grid of models with $q = 0.85$, $\varepsilon = 0.1$ and $f_{\mathrm{beta}} = 0.3$.

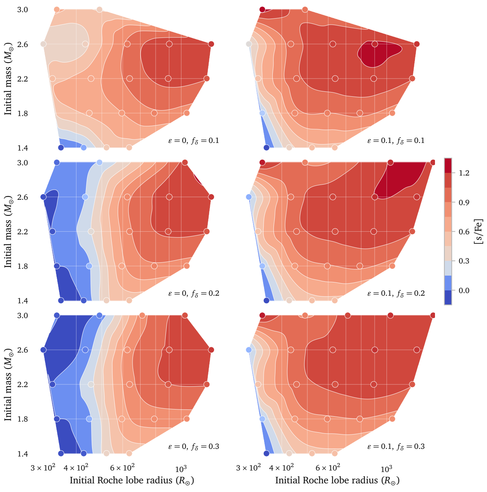
For the $f_\varepsilon = 0.3$ models, we see that the discrepancy between the two methods is much less obvious. We see a slight shift from pretty much vertical contour lines to slightly diagonal contour lines.

Obviously, for the wind only models, there is no difference between the $f_\varepsilon = 0.3$ and $f_\varepsilon = 0.1$ models.

Now, obviously I have only shown $q = 0.85$. However, the trends still exist for different mass ratios. In the appendix, I show $q = 0.7$ as well.

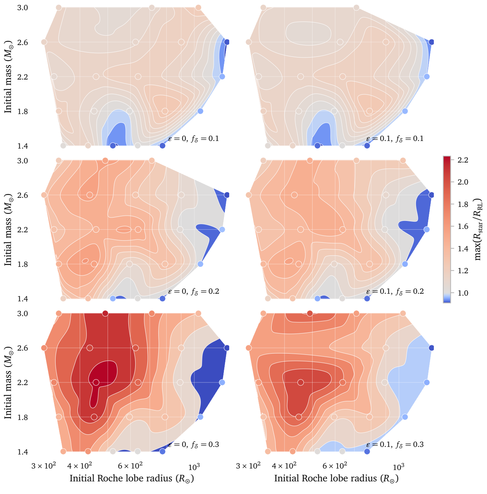
Another exciting thing is that I can probably make a much nicer version of this plot when my new grid finishes simulating.

Let me show the [s/Fe] abundances for $q = 0.7$.

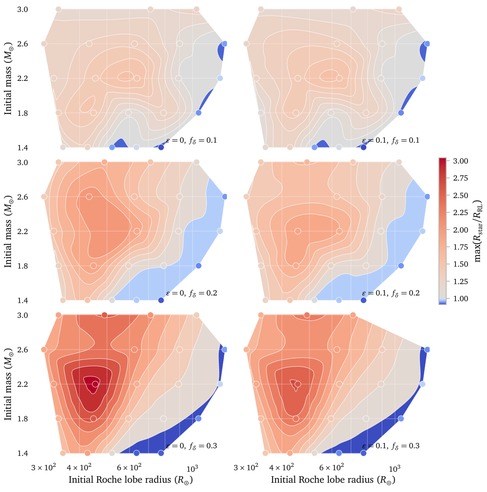


I think this looks great. Again, we could use the simple binary simulation to extend the models to higher initial Roche lobe radii, however, this is really not that relevant. We know these models with very gradually produce lower accreted abundances. This means we can easily create barium stars with wind alone, but this is not what this project is about.

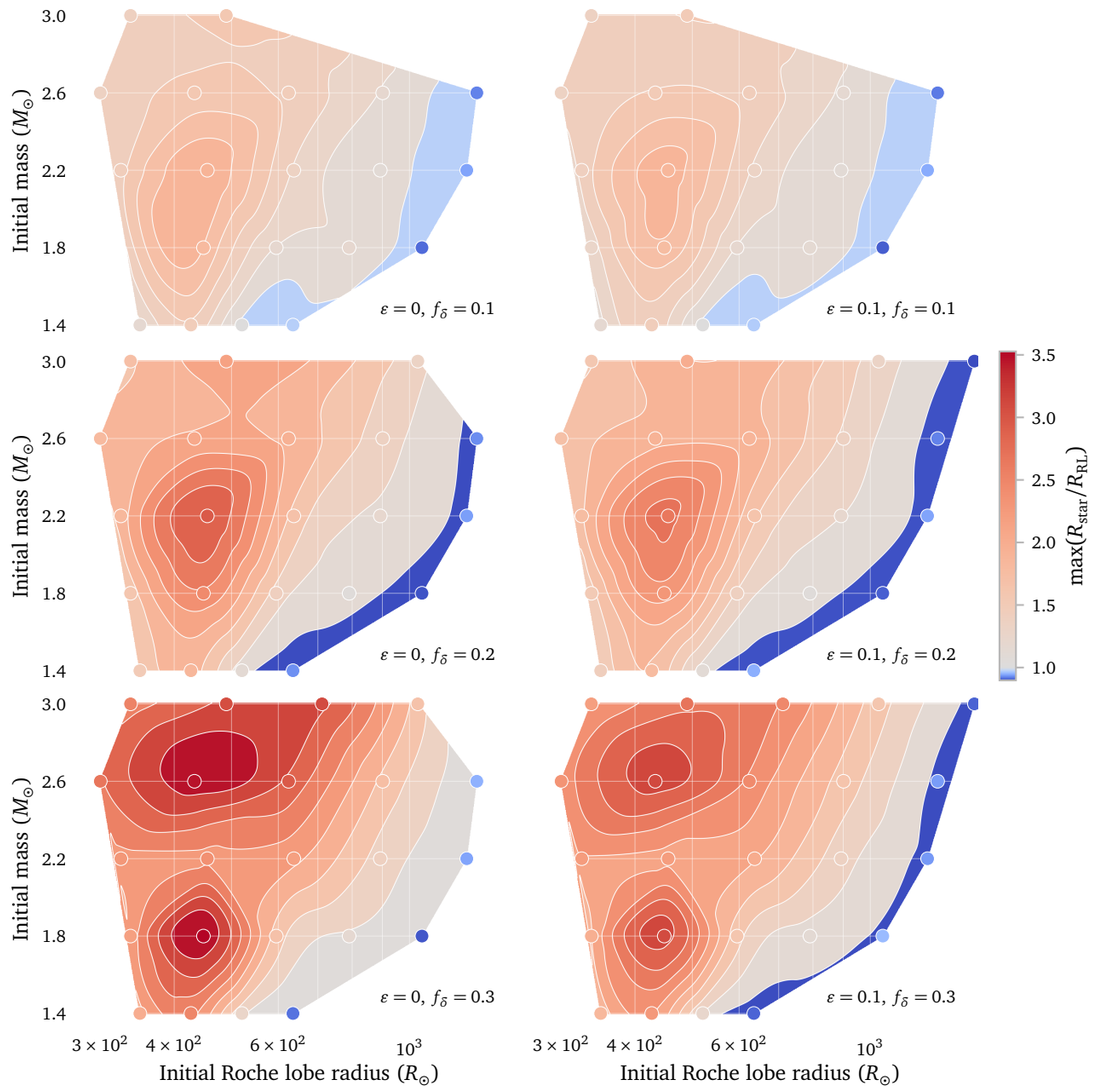
Now, let me show the overfill ratios. First for $q = 1$.

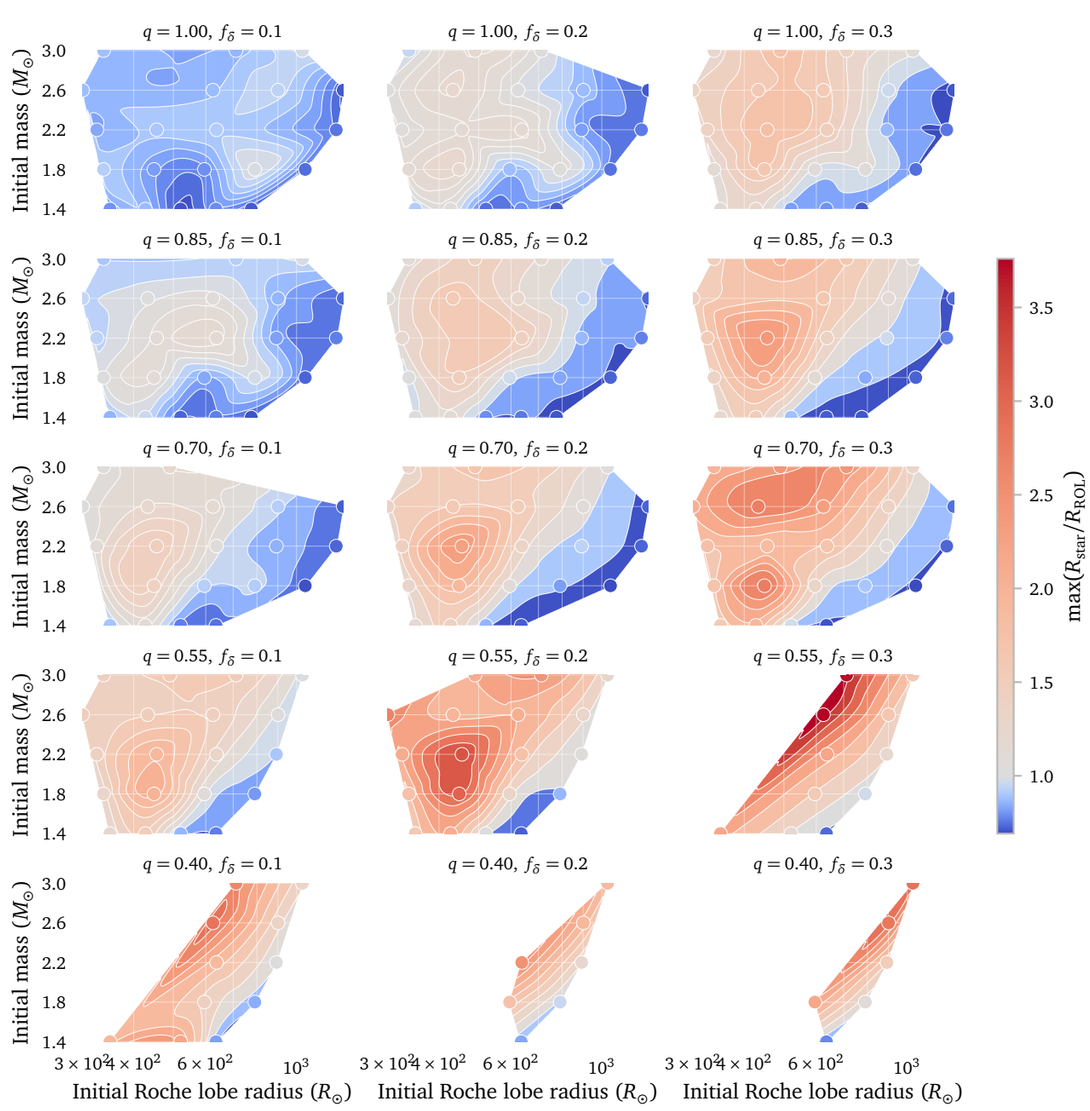


The plot below shows the same, but now for $q = 0.85$.



Lastly, for $q = 0.7$.





c. SLURM fix

I really dislike how models that timed out will not clean up their files and transfer them back to my project directory. This is really annoying for the single stars, because nearly all of them take more than 2 days to complete and I have to manually move them all back from the scratch directory to my project directory.

I did some searching on the internet and found this nice fix:

```
#!/bin/bash

# job name
#SBATCH --job-name=example

# replace this by your account
#SBATCH --account=YourAccount

#SBATCH --qos=devel
#SBATCH --ntasks=1
## Note: On Saga, you will also have to specify --mem-per-cpu

# we give this job 4 minutes
#SBATCH --time=0-00:04:00

# asks Slurm to send the USR1 signal 120 seconds before end of the time limit
#SBATCH --signal=B:USR1@120

# define the handler function
# note that this is not executed here, but rather
# when the associated signal is sent
your_cleanup_function()
{
    echo "function your_cleanup_function called at $(date)"
    # do whatever cleanup you want here
}

# call your_cleanup_function once we receive USR1 signal
trap 'your_cleanup_function' USR1

echo "starting calculation at $(date)"

# the calculation "computes" (in this case sleeps) for 1000 seconds
# but we asked slurm only for 240 seconds so it will not finish
# the "&" after the compute step and "wait" are important
sleep 1000 &
wait
```

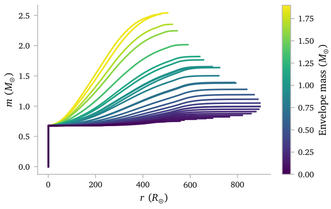
I modified this for my bash script and I now have a nice cleanup function as well that gets triggered by the USR1 trap.

I have tested this new script and it works!

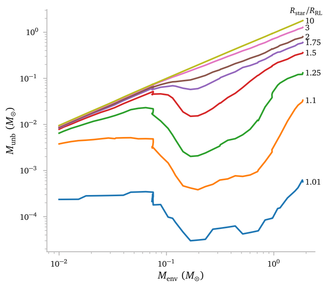
Unfortunately though, I will still need to *manually* move roughly 15 or so single star models from the scratch directories back to the project directory.

d. Unbounded mass

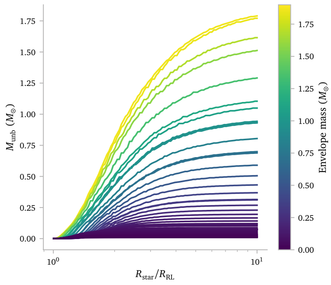
During my talk with Davey, I explained that many of my MESA models experience overfill ratios much larger than unity. This really concerned him because a lot of the *volume* of the star would then be unbounded. However, since our TPAGB stars are very puffy, I think a more interesting question is how much *mass* is actually unbounded.

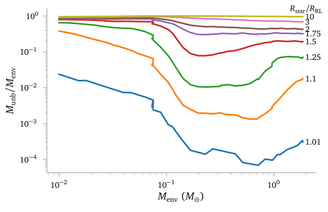


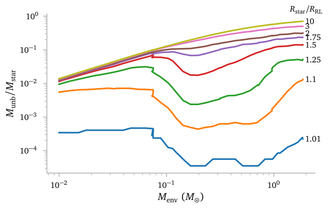
Let me start by just plotting the stellar mass as a function of the radius. It is already very clear from this picture that the core and envelope mass are pretty much distinct from each other. It is also obvious that large radii can easily occur for low envelope masses.

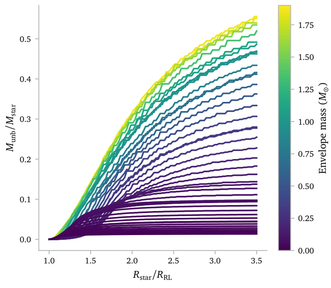


We can also plot this another way, where I now show the envelope mass as the color and the overfill ratio as the x-axis.

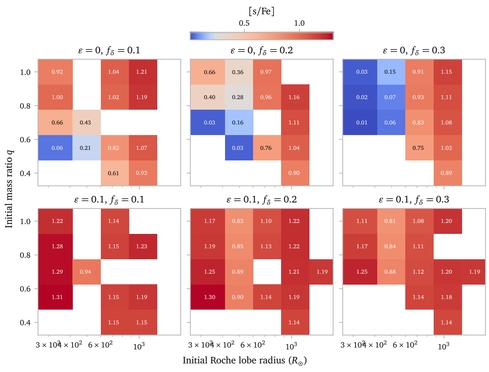






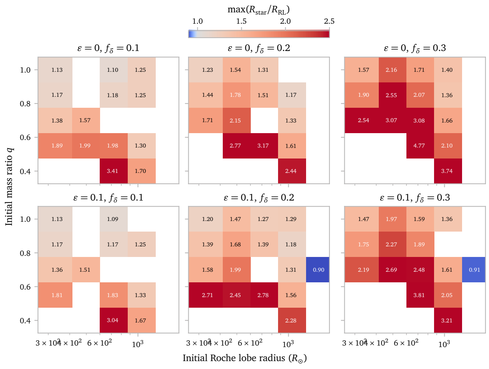


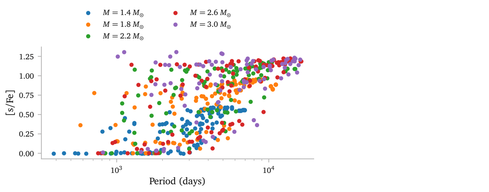
e. 3 Solar mass models



The features here look a bit unintuitive at first. However, they do make sense. The weird effects really only happen in the first two columns of every plot. We see that in many cases, the first column has a higher final abundance than the second. This is a weird effect that is caused by partial envelope stripping in the lower R models, which then causes the dredge up to more efficient in enriching the envelope, which then in turn causes wind mass transfer and subsequent RLOF to be more effective in transferring s-process abundances.

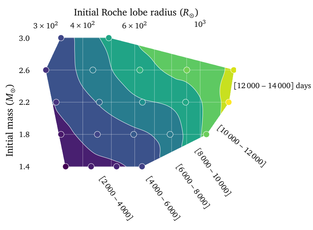
For the same reason, the $f_\delta = 0.3$ models have the lowest s-process abundances. These models often strip the entire envelope in one RLOF episode, not leaving anything to be transferred or enriched afterwards.



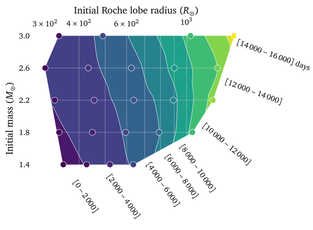


References

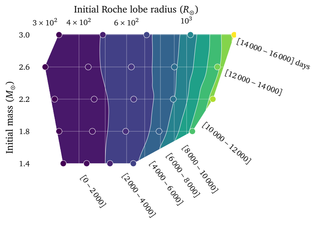
A Additional plots



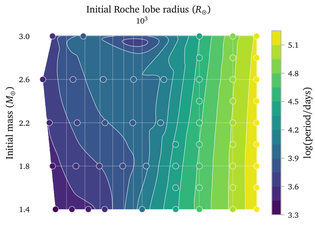
This plot shows the final period of a grid of models with $q = 0.85$, $\varepsilon = 0.1$ and $f_{\mathrm{delta}} = 0.1$.



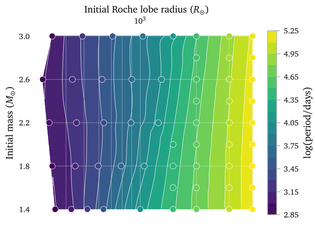
This plot shows the final period of a grid of models with $q = 0.85$, $\varepsilon = 0.1$ and $f_{\mathrm{delta}} = 0.2$.



This plot shows the final period of a grid of models with $q = 0.85$, $\varepsilon = 0.1$ and $f_{\mathrm{delta}} = 0.3$.



This plot shows a much broader plane of the final period of a grid of models with $q = 0.7$, $\varepsilon = 0.1$ and $f_{\mathrm{delta}} = 0.1$.



This plot shows a much broader plane of the final period of a grid of models with $q = 0.7$, $\varepsilon = 0.1$ and $f_{\mathrm{delta}} = 0.3$.